

Picking & Packing

Picking and packing is the process of locating and pulling product from inventory and packing it into shipment containers to fill a customer order. In this section, best practice attributes for the following process groups are covered:

- Strategy and Methods
- Tactics and Equipment
- Pick Documents
- Transactions
- Performance

Strategy and Methods

Assess your order profiles. The picking process should support both customer order profiles and product activity profiles. Typically, customer order profiles, or the way customers order your product, fall into three categories:

Order Mix: Understanding both the mix of high and low volume product and the percentage of order lines that use full pallets, full cartons, broken cartons, single units or some combination of these, allows you to plan pick area layouts and staffing.

Order Size: Understanding the number of units that customers typically order allows you to set carton size or to encourage, through marketing programs and pricing, customers to order in full carton quantities.

Order Lines: Understanding the number of lines in each order is important to setting picking strategies. A warehouse with mostly single line orders vs. one with mostly multi-line orders will have significantly different pick strategies.

Understanding product activity facilitates pick area layouts. It is important to review the order or pick frequency, how many times the product is picked, and the order volume, how much of the product is picked. Most will find their orders follow the 80/20 rule, where 80% of the orders are made up of 20% of the product stockkeeping units (SKUs). By identifying the top 20% of products, you can define the correct picking strategy. As your product mix changes you should review the impact on your picking strategy.

With an awareness of your product profile, an appropriate pick method or methods can be selected. Some common order management or order release methods are:

Single Order Picking: The most common pick method is to pick to a single order; in this process the entire order is picked. The order may be issued to the pick area as a printed-to-a-pick list, a pick label or dropped to an RF (radio frequency) terminal. The entire order may be picked and placed directly into the shipping container, eliminating downstream handling. Typically, orders are prioritized by customer-requested ship date.

Multi-Order Batch Picking: Batch processes are most effective when operators have to travel long distances in the pick area. By batching a number of orders together, a picker can pull the product for a number of orders as they pass by the product's stocking location. Batch picking works best when you have a large number of product SKUs and the products are located across a large area.

Order Consolidation: This method groups orders by destination or by customer. This has the benefit of pulling product for multiple orders in a single pass through the pick area.

Wave: A wave is an automated grouping of orders by a specific set of criteria so that these orders are released to the pick area as a group. Grouping might be based on criteria such as priority orders, orders by freight carrier, order by shipment type or orders for a specific store, location or customer. The purpose is to break the day's work into manageable segments.

Zone: Orders may be grouped by warehouse zone, such as single unit pick area, case pick area or bulk or pallet pick areas. In this case, an order may be split and consolidated in the shipping area.

Many companies use a number of order release strategies. Most WMS systems will support the above methods of order release.

Tactics and Equipment

Picking product tends to be the most labor-intensive operation in the warehouse. It is important to manage the flow of orders in the pick area to manage congestion and bottlenecks to optimize labor utilization. When laying out the pick area, seek to eliminate non-value added steps and excess travel distances. Think about product placement: place more frequently picked product in the easiest to reach locations, between the waist and shoulder in pick racks and shelving (called the "golden zone"), use storage fixtures when warranted to properly present product to pickers and locate odd-shaped, bulky or heavy product in locations that facilitate safe handling. Proper ergonomics reduce operator fatigue and injury.

The design of the pick line should support the type and volume of product you fulfill for your customers. Common examples of pick lines:

Straight Line: The pick shelving is laid out in a straight line with the most popular products placed at the front of the line to reduce the travel distance for the pick operator.

Branch and Pick Zone: In this layout there is a center aisle or conveyor and the pick shelving is placed at right angles to the center aisle. The most popular items are placed closest to the center aisle.

Serpentine Line: The operator walks from row-to-row, front to back until the order is picked completely. The most popular items are placed on the front rows closest to shipping with the least popular items on the back rows.

Pick to Conveyor: There are a number of variations with this layout, but essentially an order is passed down the conveyor from operator to operator, the operator will pick product in their pick zone and pass the order along until the entire order is completed.

Pick-to-Light: Pick-to-light systems position lights on shelving and pick racks. These lights signal the location and often the quantity of product to be picked for an order. This mix of automation and human interface greatly increases pick efficiencies and accuracy.

AS/AR (Automatic Storage and Retrieval, Automated Conveyors and Automated Carousels): There are a number of automated or semi-automated systems that hold and move product to the operator when a customer order is entered. These systems move product to the operator so that it can be packed and shipped to the customer order.

Striking a balance between manual and automated systems is the goal. It is dependant on volume, throughput requirements, order mix and available capital.

Pick Documents

How a pick task is presented to the operator varies from company to company. Technology is playing an important role in changing how orders are managed in the pick operation. Some common methods used in the warehouse are:

Paper Pick Tickets are the most common form of pick documentation. The order picker uses the document to determine the pick location and quantity. The operator must also verify that correct parts are picked and perform transactions. Paper pick tickets work well for many fulfillment operations, but are prone to human error.

Label Pick Documents work well in single order pick environments. They are a form of paper pick ticket and are used in a similar way. The added benefit is that labels are printed as part of the order release; the operator uses the label as the pick document. When the order is picked into the shipping carton, it can be packed and the labels applied. The product is ready for shipping eliminating any downstream packing steps.

Hand Held RF Terminals and Portable Label Printers are used by many warehouses to aid the picking process. A pick task is sent to the RF device and the operator picks the product. The product is validated by scanning the product barcode ID or RFID tag. At the end of the task, a shipping or consolidation label may be printed either on a portable printer or within the work area. The process can be virtually paperless.

Pick-to-light allows for paperless picking, when pick-to-light equipment is linked to the order management and inventory system. The operator picks product based on the lighted location and then confirms each pick in the system. The system is then able to carry out inventory transactions, complete order records and drive replenishment requirements.

Voice recognition technology sends voice messages to operators and allows operators to use common speech to give commands to the system. Pick tasks are delivered to the operator and the system helps to direct the operator to the pick location. Voice systems are flexible and allow for order priorities to be quickly changed. Many companies have set up dual systems, using pick-to-light for the 20% of the items that make up 80% of the product volume and voice for the 80% of the parts that make up 20% of the product volume.

There are a number of methods available to manage the picking process, selecting the correct one for your operation will help you achieve operational efficiency. And remember to factor in ergonomic considerations.

Transactions

Technology plays a part in a company's ability to perform transactions as a seamless part of the picking process. It is common practice for transactions to be performed at the end of the pick process as part of order consolidation or confirmation. Best practice companies have integrated transactions into the process by using RF terminals, wireless speech system or similar WMS system enabled transaction automation. This allows transactions to take place in real time and in a single system of record.

Performance

Performance metrics must measure both what the customer sees and what drives improvement in warehouse processes. Good metrics are ones that are linked to customer satisfaction, are documented with operations definitions, and that drive improvement. Good and best practice companies will:

- Measure daily activity by major task
- Measure individual accuracy and performance
- Display performance metrics on the warehouse floor
- Include employees in continuous improvement programs
- Gather feedback, suggestions and information from the operators
- Train and cross train
- Report metrics to customers

Improving performance in the pick area will help to reduce labor and increase efficiency; it will also boost customer service levels.

STEP 1. Picking & Packing Process Benchmarks

Review this chart and consider your pick and pack processes as they relate to the listed attributes. Check or highlight the description that sounds most like your operation.

Pick and Pack – Process Benchmarks					
Process Group	Poor Practice	Inadequate Practice	Common Practice	Good Practice	Best Practice
Strategy and Methods	No picking strategy No review process of pick methods	Picking strategy is not adequate to support current customer requirements Process to review pick methods not well defined	Picking strategy supports current customer requirements and may include more than one pick/pack process Review pick processes and strategy by product group typically once per year Simulations and modeling may be used on a project basis	Picking strategy supports current customer requirements and may include a number of pick/pack processes (Zone, Wave, Batch/consolidation, Pick and pass, Single order pick, Kit picks) Review pick processes and strategy for each product at least once per year Modeling and simulations may be performed as part of review process	Picking strategy supports current and forecasted customer requirements and will include multiple optimized pick/pack processes Optimized wave picking and task interweaving Review pick processes and strategy for each product at least once per quarter Modeling and simulations are run frequently
Tactics and Equipment	Pick/pack areas are not arranged to support current demand, issues meeting current volume Little or no specialized material handling equipment Operator efficiency not managed Poor ergonomics of pick areas, issues with operator fatigue and injury Poor housekeeping	Pick areas are not arranged to support current demand Little specialized material handling equipment used Operator pick efficiency is monitored but not managed well Pick/pack areas are not laid out well, poor ergonomics, issues with operator fatigue Moderate housekeeping, obvious safety issues	Pick areas are laid out well to support current demand Some specialized material handling equipment is used to improve efficacy Operator pick efficiency is monitored and managed Pick/pack areas are laid out well, good ergonomics, operator fatigue minimized Good housekeeping, some safety issues	Pick areas are laid out well to support current and peak demand Material handling equipment used to improve efficacy and reduce travel time (Pick to light, carousels, flow rack may be used) Operator pick efficiency and travel time is monitored and managed Pick/pack areas laid out ergonomically with the objective of reducing employee fatigue and injury Good housekeeping	Pick areas are optimized to support current and future demand Conveyors or other automated material handling equipment to bring the orders into each required pick zone, eliminating travel time for pickers (Pick to Light, AR/AS, flow rack, auto pick equipment, may be used) Operator pick efficiency and travel time is monitored and optimized All pick/pack areas laid out ergonomically to reduce employee fatigue and injury Excellent housekeeping

Pick and Pack – Process Benchmarks					
Process Group	Poor Practice	Inadequate Practice	Common Practice	Good Practice	Best Practice
Pick Documents	Manual and paper based pick lists Pick jobs not well managed	Pick documents are not sequenced No consistent process for job release	Pick documents support limited sequencing based on locations Jobs are re-leased in batches	Pick process supports pick sequencing based on travel path or locations Release of pick jobs in waves, documentation supported by RF	Pick travel path minimization through order picking in travel path sequence Batch picking of the same SKUs for multiple orders, or wave pick sequencing to plan picks per zone in advance
Transactions	Stationary terminals used in pick areas	Stationary terminals used in pick areas Systems do not support transaction process well, possible multiple systems with batch processing	Mix of RF and stationary terminals used in pick areas Transactions may be batch processed Multiple integrated systems may be used, transactions are integrated among systems	Pick areas use RF terminals in combination with portable printers and stationary terminals and printers Transactions are in near real time If multiple systems are used there is seamless system integration among systems Systems support RFID tag / Electronic Product Code tracking when required	RF terminals, wireless speech system or similar WMS system enables automated order communication to personnel, portable printers used Transactions are in real time Single system of record RFID tag / Electronic Product Code tracking integrated into pick process when required
Performance	Inconsistent record of activity by major task No measures of staff productivity Customers are not included in the performance review process	Record of monthly activity by major task and manning levels held by manager Measure shift performance Customers may receive performance information	Record of weekly activity by major task and manning levels held by manager Measure shift productivity performance Customer receive quarterly performance reports	Record of weekly activity by major task and staffing levels displayed on warehouse floor. Measure shift or individual productivity performance Customers receive or have access to performance reports	Record of daily activity by major task and staffing levels displayed on warehouse floor. Employees are included in continuous improvement programs Productivity targets set and measured, showing an improving trend and / or meeting goals Customers can review performance activity level via on-line reporting
Supply Chain Visions- Best Practice Process Attributes and Benchmarks (Copyright 2007 Supply Chain Visions)					

STEP 2. Assessment for Picking & Packing

Directions: Rank your processes against the Picking and Packing Benchmarks in STEP 1. Use this chart to check which column best represents your processes. In some cases, you may find that you fall between rankings and you can add 0.5 to the lower rank. Once each process is assessed, total the values for the group.

Pick and Pack – Self-Assessment					
Process Group	Poor Practice	Inadequate Practice	Common Practice	Good Practice	Best Practice
Rank	1	2	3	4	5
Strategy and Methods					
Tactics and Equipment					
Pick Documents					
Transactions					
Performance					
Self Assessment Score Total					

STEP 3. Ranking Ranges for Picking & Packing

Directions: Use this table to compare your self-assessment score TOTAL to the Picking and Packing section score rankings listed here.

Picking and Packing - Ranking					
	Poor Practice	Inadequate Practice	Common Practice	Good Practice	Best Practice
Section Score Rankings	5	10	15	20	25

Our self-assessment score TOTAL of _____ means that we rank as
☐ poor ☐ inadequate ☐ common ☐ good ☐ best practice in this area.

STEP 4. Key Performance Metrics for Picking & Packing

The table below provides quantitative benchmark data on a selected number of KPIs relevant to Picking and Packing. Each KPI is scaled from *Major Opportunity* to *Best in Class*. The values are from a recent WERC Benchmarking Study.

KPI	Poor Practice	Inadequate Practice	Common Practice	Good Practice	Best Practice
	<i>Major Opportunity</i>	<i>Disadvantage</i>	<i>Average</i>	<i>Advantage</i>	<i>Best in Class</i>
On time shipments	< 94%	>=94% and <97.0%	>=97% and <98.9%	>=98.9% and <99.7%	>= 99.7%
Fill Rate - Line	< 92%	>=92% and <97%	>=97% and <98.54%	>=98.54% and <99.66%	>= 99.66%
Fill Rate - Order	< 92%	>=92% and <96%	>=96% and <98.5%	>=98.5% and <99.7%	>= 99.7%
Average time from order placement to shipment	>=60 hours	>=36 and <60	>=24 and <36	>=8 and <24	< 8
Order picking accuracy	<98%	>=98% and <99%	>=99% and <99.5%	>=99.5% and <99.9%	>= 99.9%
Percent of orders shipped complete	<92%	>=92% and <96%	>=96% and <98.5%	>=98.5% and <99.3%	>= 99.3%
On time delivery	<92%	>=92% and <95%	>=95% and <98%	>=98% and <99%	>= 99%
Perfect Order Index	<86.92%	>=86.92% and <95%	>=95% and <98%	>=98% and <99.48%	>= 99.48%
Cases shipped per person hour	<30	>=30 and <70	>=70 and <145.36	>=145.36 and <286.4	>= 286.4
Pallets shipped per person hour	<4.6	>=4.6 and <10	>=10 and <18	>=18 and <34.4	>= 34.4
Backorders as a % of total orders	> =10%	>=5% and <10%	>=2.12% and <5%	>=0.484% and <2.12%	> 0.484%

KPI Definitions		
KPI	Definition	Calculation
On time shipments	Off the dock, on the truck and in transit to its final destination.	# of orders shipped on time / # of total orders shipped. On time is defined by the customer request and SHOULD NOT be moved once it is entered into the system
Fill Rate - Line	This metric measures the percentage of line items on a specific order shipped from stock within 24 hours of order release.	# line items filled from stock within 24 hours of order release / total number of lines on the order
Fill Rate - Order	This metric measures the percentage of orders shipped from stock within 24 hours of order release.	# of orders filled from stock shipped within 24 hours of order release / total number of stock orders
Average time from order placement to shipment	The time between order placement by the customer and order shipment from the supplier.	date order shipped - date order placed
Order picking accuracy	This measures the accuracy of the orders picked. Errors may be caught prior to shipment.	orders picked correctly / total orders picked
Percent of orders shipped complete	Complete means that all lines / units ship with the order.	# of orders shipped with all lines & units / total number of orders shipped
On time delivery	The percentage of orders that arrive at their final destination on the date agreed upon.	# of orders delivered on time / total number of orders
Perfect Order Index	An order that meets all of the following criteria: complete, on-time (as defined by customer - could be on time delivery or on time shipment), perfect documentation, perfect condition.	% of on time orders x % of complete orders x % of damage free orders x % of orders with accurate documentation as defined by customer (e.g. invoice, ASN, labels)
Cases shipped per person	The number of individual cases of product shipped by an employee.	# of cases shipped by employee / total number of cases shipped
Pallets shipped per person	The number of pallets of product shipped by an employee	# of pallets shipped by employee / total number of pallets shipped
Backorders as a % of total orders	The portion of total orders that are held and shipped late due to lack of availability of stock. Can be measured by lines or by PO, by units or by \$ value.	# or \$ of orders (or lines or units) held and not shipped / total # or \$ of orders (lines or units)
WERC Benchmarking Study 2007 (Copyright 2007 WERC/Georgia Southern University/Supply Chain Visions)		

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Case Study: Talking About an Evolution

Wireless technology is giving DCs a big boost to productivity

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While everyone is busy talking about the many benefits of radio frequency identification (RFID), there's a quieter evolution going on with other wireless products in the DC. While RF products like handheld computers, barcoding and voice technology may not be in the limelight, their role in the DC is critical. You might even say wireless technology is rapidly becoming the backbone of DC operations.

"Just about every DC today has some sort of RF network," says Bill Hubacek, director of distribution technologies at FKI Logistex North America, a global provider of automated material handling solutions located in St. Louis, Mo. "There's a need for real-time information, and these tools can deliver that."

Mark Dessommes, marketing director at LXE, Inc., a wireless product solution company based in Norcross, Ga., says that wireless tools are key to increased productivity. "Your employees can work faster because they are working all the time, rather than coming and going," he says. "You can also expand your business without expanding your workforce."

Other benefits from wireless technology, according to Dessommes, include:

- Increased inventory control, leading to the ability to do more with less space
- Improved customer service
- Reduced processing/shipping time a Real-time traceability
- The ability to provide customers with real-time visibility into their orders

Handheld computers, barcode scanners, voice technology and imaging technology have all improved greatly over the past 5 to 10 years. "Today you get both reliability and speed," says Hubacek.

Not only that, but going wireless today is more affordable than ever. "It's not so much that the price has come down, but that the capability of these tools has gone up," says Hubacek. "You get so much more from today's technology than you did just 10 years ago, and you get it for the same price."

Tools that deliver

"You can find handheld computers, barcoding, image capture and other tools that can help track the movement of inventory in and out of the DC," says Richard Baully, vice president of strategy and business development at PsionTeklogix, a global provider of mobile computing solutions located in Mississauga, Ontario, Canada. "They are more capable than ever and allow you to do more than in the past."

For instance, today's handheld computers frequently come equipped with graphic interfaces, providing "a laptop in your hand," says Baully. Rather than simply sending a text message to a terminal instructing a picker on where to go, these terminals can now display complex graphics and screens that could be found only on PCs just a few years ago.

Baully says that software is much more "mobile aware" than in the past. "Now these tools can interact more efficiently with the back-end evolution," he says.

That means that the architecture is now more like a wired local area network (LAN), giving users the ability to connect with other supply chain management areas. Look for wireless devices to soon plug into applications other than just WMS, such as an enterprise resource planning (ERP) system. This adds more value to the technology investment, making it even easier to justify.

Dual-use tools

Dual-use tools are also making an impact. Baully says that image capture technology, which is often found in a tool that also provides scanning capability, is playing an increasingly bigger role in the DC. With tools that both scan and capture images, "You can take pictures of envelopes, inventory or damaged goods and send them to the server or save them for later," he says.

"We're just starting to see more dual-use technologies," says Dessommes, "and there's definitely interest in it." Another example is wireless devices that can provide voice technology and scanning capabilities.

Voice technology itself is becoming an ever more common wireless tool in the DC. Using RF infrastructures, some companies have installed voice communication systems that allow pickers and putters in the DC to receive and acknowledge instructions through headsets instead of screen-based directions. "Speech technology has the advantage of freeing up your employees' hands so that they can be more productive," says Baully.

The future

A peek into the crystal ball reveals that tomorrow's wireless tools promise even more. Baully says to expect to see a "convergence of technology." Hubacek agrees. "There's going to be a continuation of the wireless computing concept that will take the technology to the next level," he says.

"Once companies have the backbone in place, the devices will be there and they will find more and more applications for wireless," he says. "You'll see a network within a network where wireless devices will be able to communicate with each other."

Dessommes points to several specific wireless applications that are on the horizon. The first is WiMax - high-speed, wireless communication technology. "This is being developed to solve problems in the WAN world," he says. "It will have a role in outdoor storage applications."

Another is what he calls "mesh networking." This is a scenario where wireless devices are able to use each other for connecting back to the host. "Expect to see mesh networking options for logistics applications sometime in 2006," he says.

Also look for open standards, which are rapidly becoming reality. The advantage is that wireless devices from different vendors can communicate with one another. This allows companies to piece together the ideal system and tools for their needs rather than having to make their processes fit the technology.

Interconnectivity between wired and wireless tools is also improving, making for better tracking of processes on the DC floor. A pick-to-light system combined with voice technology is a good example of this new technology "relationship."

Finally, all of the vendors say that wireless tools are becoming increasingly advanced and that will lead to an even faster return on your investment, something that is already usually in the six- to nine-month range, according to Dessommes. "All of the tier one DCs have gone wireless," he says, "and now the small DCs are starting to get there as well."

And that's something that, today, can't be said about RFID.

Making it happen – an example

Everyone in the warehouse or DC understands the importance of speed—customers want their products yesterday, so there's no room for error. No one knows this better

than Martin Bros. Distributing Co., a full-line foodservice distributor that delivers products to foodservice establishments, schools and care facilities in the Midwest and foodservice and restaurant equipment nationwide.

With over 2,500 customers, Martin Bros moves approximately 150,000 items through its DC each week. To meet customer demands, the company cannot afford miss-picks, missing items or delays. Recognizing this, the company decided that implementing wireless systems was necessary. Martin Bros. focused in on automating the data capture process of goods moving in, through and out of its DC.

The company decided on a variety of wireless solutions from Symbol Technologies, Inc. For receiving and storage, they selected legacy mobility solutions that include a portable data terminal designed for rugged environments, along with Symbol's forklift-mounted reader. Both products move data over frequency hopping (FH) wireless networks. In addition, Martin Bros. added a wearable scanning system that could receive stock picking and inventory management information over Wi-Fi networks.

This required that Symbol add a wireless switch to centralize the DC's entire wireless network. Symbol and its partner, Systems Application Engineering (SAE), then deployed the new mobile Wi-Fi shipping system using the Symbol wearable computer and data capture system. The switch technology allowed Martin Bros. to preserve old applications while adding new and to run them over the same wireless network. The distributor now has a centralized wireless switch architecture with a fully functioning FH and Wi-Fi wireless network infrastructure that supports all necessary mobile applications.

Best of all, Martin Bros. was able to reduce stock picking errors and increase productivity, resulting in lower administration costs and improved customer satisfaction.